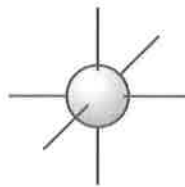
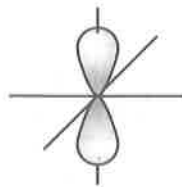


- 4 The diagrams show the shape of two orbitals, with principal quantum number 3, in an isolated atom.



X



Y

Which orbital has the lower energy and how many of this type of orbital have the principal quantum number 3 in a single isolated atom?

	orbital with lower energy	number of orbitals of that type with principal quantum number 3
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